Received: 5 January 2026 | Revised: 18 April 2026 | Accepted: 21 April 2026
DOI: 10.1111/jfb.70486

REGULAR ARTICLE

JOURNAL OF FISH BIOLOGY

Conservation genomics of Pecos pupfish (*Cyprinodon pecosensis*)

Elizabeth P. Dolan[1] | **Andrew T. Fields**[1] | **Megan G. Bean**[2] | **Robyn R. Doege**[3] | **Joanna L. Hatt**[4] | **Kevin W. Conway**[5] | **David S. Portnoy**[1]

[1]Marine Genomics Laboratory, Department of Life Sciences, Texas A&M University-Corpus Christi, Corpus Christi, Texas, USA

[2]U.S. Fish and Wildlife Service, Austin Ecological Services Field Office, Austin, Texas, USA

[3]Fort Worth Zoo, Fort Worth, Texas, USA

[4]New Mexico Department of Wildlife, Santa Fe, New Mexico, USA

[5]Department of Ecology and Conservation Biology and Biodiversity Research and Teaching Collections, Texas A&M University, College Station, Texas, USA

Correspondence
Elizabeth P. Dolan and David S. Portnoy, Texas A&M University-Corpus Christi, 6300 Ocean Drive, Corpus Christi, Texas 78412, USA.
Email: elizabeth.dolan@tamucc.edu and
Email: david.portnoy@tamucc.edu

Funding information
Texas Parks and Wildlife Department, Grant/Award Number: F21AP03616

[Correction added on 19 May 2026, after first online publication: The middle initial of the fourth author has been corrected from 'L.' to 'R.,' so it now reads Robyn R. Doege throughout the article.]

Abstract

The Pecos pupfish, *Cyprinodon pecosensis*, is an imperilled freshwater fish found in arid regions of Texas and New Mexico (USA). The species faces multiple challenges to persistence including reductions in suitable habitat, water shortages, as well as hybridization and competition with an introduced congener (sheepshead minnow, *C. variegatus*). As part of the current management strategy, refuge populations, seeded with non-introgressed individuals collected from Texas, are maintained at the Fort Worth Zoo and on private property in West Texas. Therefore, assessments of standing genetic variation within and among wild and refuge populations and levels of admixture in the wild are critical for future conservation and management planning. Fin clips were acquired in 10 locations in the wild, five in Texas, five in New Mexico and from two refuge populations in Texas (one maintained by Fort Worth Zoo and another on a private property in West Texas). In Texas, non-introgressed *C. pecosensis* were found in the wild at only one location Upper Salt Creek (USC); all other locations were composed of admixed individuals or non-introgressed *C. variegatus*. No admixed individuals were found in New Mexico. Significant genetic heterogeneity was detected between all locations of non-introgressed *C. pecosensis*, including the refuge populations, and estimates of divergence between Texas locations (USC and refuge populations) and New Mexico locations were relatively large (F_{ST}: 0.23–0.32). Estimates of contemporary effective population size for USC and the two refuge populations were less than 50, but greater than 500 for all New Mexico populations. In summation, while New Mexico populations look secure, the data suggest that *C. pecosensis* in Texas is currently imperilled by the threat of hybridization, as well as potential loss of genetic diversity on contemporary time scales. Deep divergence between Texas (including refuge populations) and New Mexico suggests that further study would be needed prior to any management plans that would involve assisted gene flow between the regions.

KEYWORDS
Cyprinodontidae, desert fishes, genetic rescue, genetic swamping

1 | INTRODUCTION

North America is home to nearly 1200 native freshwater fishes across 50 families and 201 genera, accounting for roughly 10% of the global freshwater fish fauna (Page and Burr, 2011; Warren Jr. & Burr, 2015). Many of these species occur in the southeastern United States, which has a high level of species diversity and endemism due to favourable climate and stable habitats over recent geological time, as well as high historical rates of colonization and speciation (Smith et al., 2010). By contrast, the southwestern United States, has low levels of species diversity due to challenging environmental conditions and high rates of extinction over recent geological time, with many species comprising a series of fragmented and relic populations (Smith et al., 2010). In addition to the inherent difficulties of living in the arid southwestern United States, many of these fishes also face challenges due to anthropogenic factors such as habitat alteration, water extraction and the introduction of non-native species (Jelks et al., 2008). These mounting pressures have led to many fishes in the region being state and federally listed as threatened or endangered, with some species recently extirpated from parts of historical ranges, or in some cases driven to extinction (e.g., Calamusso et al., 2005; Jelks et al., 2008; Moyle & Leidy, 1992; Olden & Poff, 2005; Osborne et al., 2021).

One such impacted system is the Pecos River, which originates near Pecos, New Mexico and ends at its confluence with the Rio Grande River near Langtry, Texas. The upper reaches of the Pecos River are primarily sustained by seasonal snow melt, while the lower reaches depend on water runoff during monsoon rains during the warm season and spring-fed tributaries with highly variable seasonal flow rates (Yuan et al., 2007). There are many natural and anthropogenic stressors throughout the Pecos watershed such as drought, increasing temperature, water removal for human activities, and the introduction of non-native species. These pressures have led to a decrease in water availability and quality, decreased flow rates, increased salinity, and increased competition, predation and hybridization—factors that have had profound impacts on the native fish community (Cheek & Taylor, 2016; Harley & Maxwell, 2018; Jensen et al., 2006; Yuan et al., 2007). Historic stream capture events shaped the ichthyofauna of the Pecos watershed; and although not highly diverse, it supports a variety of species of conservation concern (Craig & Bonner, 2019; Hoagstrom et al., 2025; Lee et al., 1980; Perkin et al., 2021), including two endemic roundnose minnows (*Dionda episcopa* Girard, 1856 and *Dionda* sp. 'Upper Pecos' Schönhuth et al., 2012), three endemic gambusia (*Gambusia echelleorum* Portnoy et al., 2025, *Gambusia nobilis* Baird & Girard, 1853, and *Gambusia pyrros* Portnoy et al., 2025) and three endemic pupfishes (*Cyprinodon bovinus* Baird & Girard, 1853, *Cyprinodon elegans* Baird & Girard, 1853, and *Cyprinodon pecosensis* Echelle & Echelle, 1978).

Endemic pupfishes of the genus *Cyprinodon* are of particular conservation concern in the southwestern United States. The genus comprises 31 named species in the southwest of the United States (Arizona, California, New Mexico, Nevada and Texas) as well as northern Mexico (Echelle & Echelle, 2020; Fricke et al., 2025). Within the US portion of the Rio Grande drainage there are four imperilled species, including: *C. bovinus* (Leon springs pupfish), *C. elegans* (Comanche springs pupfish), *Cyprinodon eximius* Girard, 1859a (Conchos pupfish) and *C. pecosensis* (Pecos pupfish). Like many other *Cyprinodon* species across the southwest, these species inhabit fragmented spring-fed habitats, which are highly susceptible to anthropogenic activities in surrounding areas (Burkhead, 2012; Elkins et al., 2024; Hoagstrom & Osborne, 2021). In addition, many *Cyprinodon* population sizes are estimated to be low, which may be a concern for long-term maintenance of adaptive potential and in some cases persistence in the absence of natural gene flow (Blomqvist et al., 2010; Hoffmann et al., 2017; Willi et al., 2006). Several species of *Cyprinodon* in the southwestern United States and Mexico have already been extirpated from the wild mainly due to habitat destruction and the pumping of groundwater leading to the loss of spring flow (Echelle & Echelle, 2020). Broader conservation efforts in the region have included land acquisition, conservation easements and habitat restoration efforts by private landowners, non-governmental organizations, as well as state and federal agencies, and these actions have benefited several species of *Cyprinodon* in Texas and New Mexico (Minckley et al., 1991). Additionally, the widespread introduction of the non-native sheepshead minnow (*Cyprinodon variegatus* Lacepède, 1803) throughout the region poses another threat because of competition and hybridization with native *Cyprinodon* species. Refuge populations have been established to preserve the integrity of these species and to be used as a safeguard, allowing for reintroduction into the wild if populations become extirpated or inundated with hybrids (Edds & Echelle, 1989).

C. pecosensis was formally described in the late 1970s (Echelle & Echelle, 1978), prior to which individuals of the species were assumed to represent *C. bovinus*, due to the poor state of the type material. Following the rediscovery of *C. bovinus* (Echelle & Miller, 1974) in Diamond Y, morphological and genetic differences between *C. bovinus* and *C. pecosensis* were identified, resulting in the description of the latter (Echelle & Echelle, 1978). *C. pecosensis* is a small, deep bodied fish, with adults ranging from 30 to 46 mm in length (Echelle & Echelle, 1978). This species is sexually dimorphic with adult males and females presenting different coloration and patterning on the body and fins (Echelle & Echelle, 1978; Figure 1). Historically, the species was reported throughout the Pecos River system and its tributaries, from Chaves County, New Mexico, south to Val Verde County, Texas (Echelle et al., 1997; Hendrickson & Cohen, 2022). However, it has experienced an 80% range contraction and now exists in fragmented populations (Echelle & Echelle, 2003; Hoagstrom et al., 2025). The species is listed as vulnerable by IUCN Red List of Threatened Species (Nature Serve, 2013) and was proposed for listing as threatened with a 4(d) rule and designation of critical habitat under the Endangered Species Act on November 22, 2024 (US Fish and Wildlife Service, 2024).

The introduction of *C. variegatus* into the Pecos River is thought to have occurred between 1980 and 1984, likely near the Red Bluff Reservoir near the New Mexico/Texas state line (Echelle & Connor, 1989). The introduction resulted in rapid hybridization

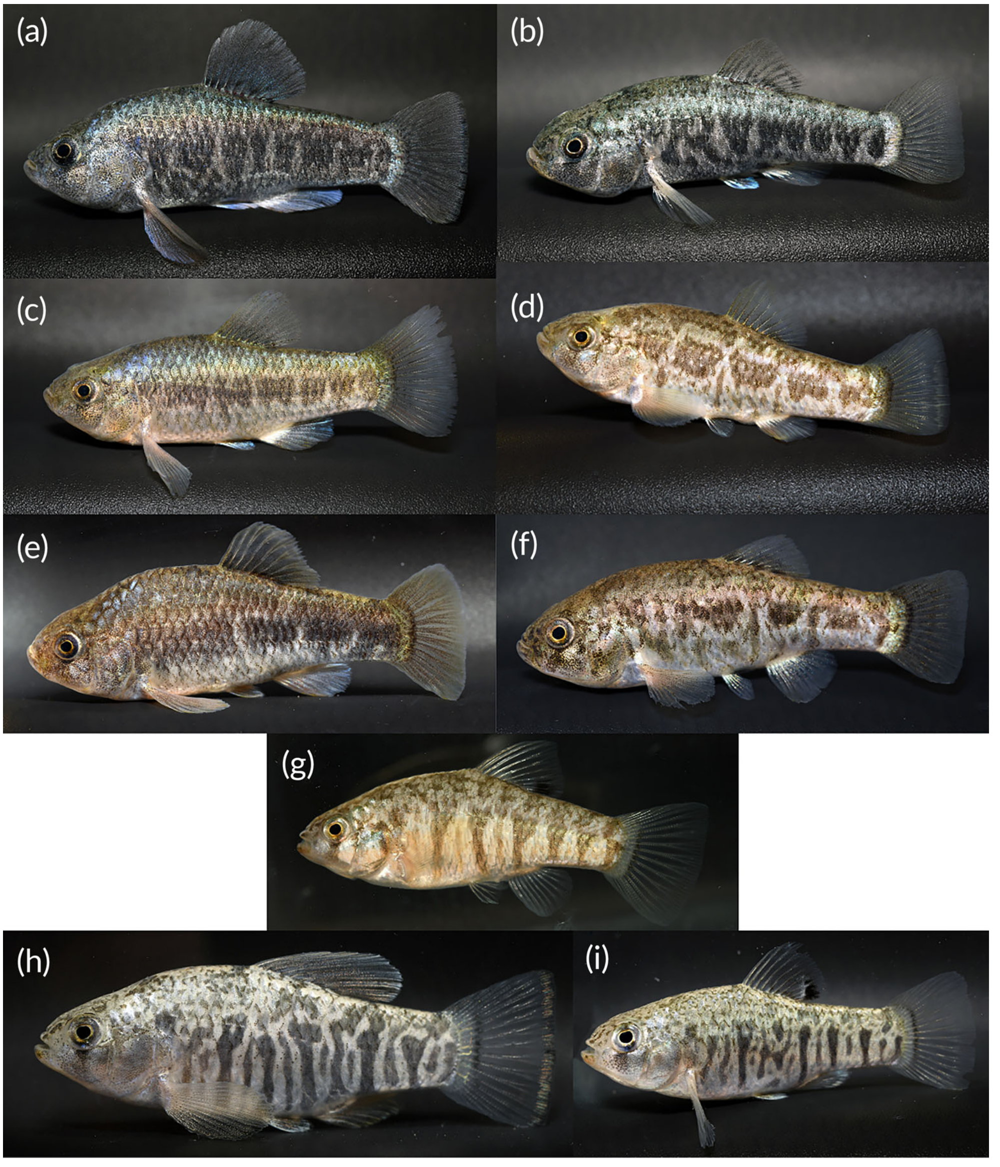

FIGURE 1 Photographs of select individuals of *Cyprinodon pecosensis* (a–f, males on left, females on right), a potential hybrid between *C. pecosensis* x *C. variegatus* (g), and *C. variegatus* (h–i, male on left, female on right). (a, b) TCWC 20594.01, private ranch, Reeves Co., TX; C, D, TCWC 20596.01, Salt Creek at private ranch, Reeves Co., TX (USC); (e) F, Bitter Lake, Chaves Co., NM (AMN), not preserved; (g) TCWC 20597.01, female, Salt Creek, Reeves Co., TX (LSC); (h, i) TCWC uncat., Balmorhea Lake, Reeves Co., TX.

between *C. pecosensis* and *C. variegatus* as multiple hybrids were confirmed in the Pecos River in Texas at sampling locations 200 river-km apart in 1984 (Echelle et al., 1987). It is hypothesized that the rapid rate of hybridization is due to the increased ecological fitness of hybrid offspring when compared to the parental species (Rosenfield et al., 2004). Currently, most of the range of *C. pecosensis* in Texas contains admixed populations of *C. pecosensis* and *C. variegatus* with a varying degree of mixing based on locality (Echelle & Connor, 1989; Echelle & Echelle, 2003). By contrast, there is little evidence of widespread hybridization in the New Mexican reaches of the Pecos River, with admixed individuals being documented only within the first 55 km of river north of the Texas border, likely due to bait bucket introductions (Echelle et al., 1997). Currently, there is no documented hybridization north of Loving Crossing, New Mexico or in isolated ponds, away from the Pecos mainstem (Echelle et al., 1997). In Texas, outside of stocked artificial ponds, it is likely that the only remaining wild population of non-introgressed *C. pecosensis* persists in Salt Creek, near the Texas-New Mexico border (Echelle & Echelle, 2003).

Since its description, *C. pecosensis* has been the focus of a considerable amount of research, spanning a variety of topics including ecology, conservation and evolution (Collyer et al., 2015; Echelle et al., 1997; Garrett et al., 2002; Kodric-Brown & Mazzolini, 1992; Kodric-Brown & Rosenfield, 2004; Rosenfield et al., 2004; Xu, 2017). Genetic studies available for *C. pecosensis* focused on understanding the genetic differences between refuge populations and their wild counterparts or evaluating the extent of hybridization between *C. pecosensis* and *C. variegatus* but used only a small number of allozymes or a portion of mitochondrial DNA (Childs et al., 1996; Echelle et al., 1997; Edds & Echelle, 1989; Wilde & Echelle, 1992). The latter is of particular concern for *C. pecosensis*, which is known to form hybrid swarms with *C. variegatus* and appears to be susceptible to extinction via hybridization (Childs et al., 1996; Garrett et al., 2002). While the hybridization between *C. pecosensis* and *C. variegatus* has

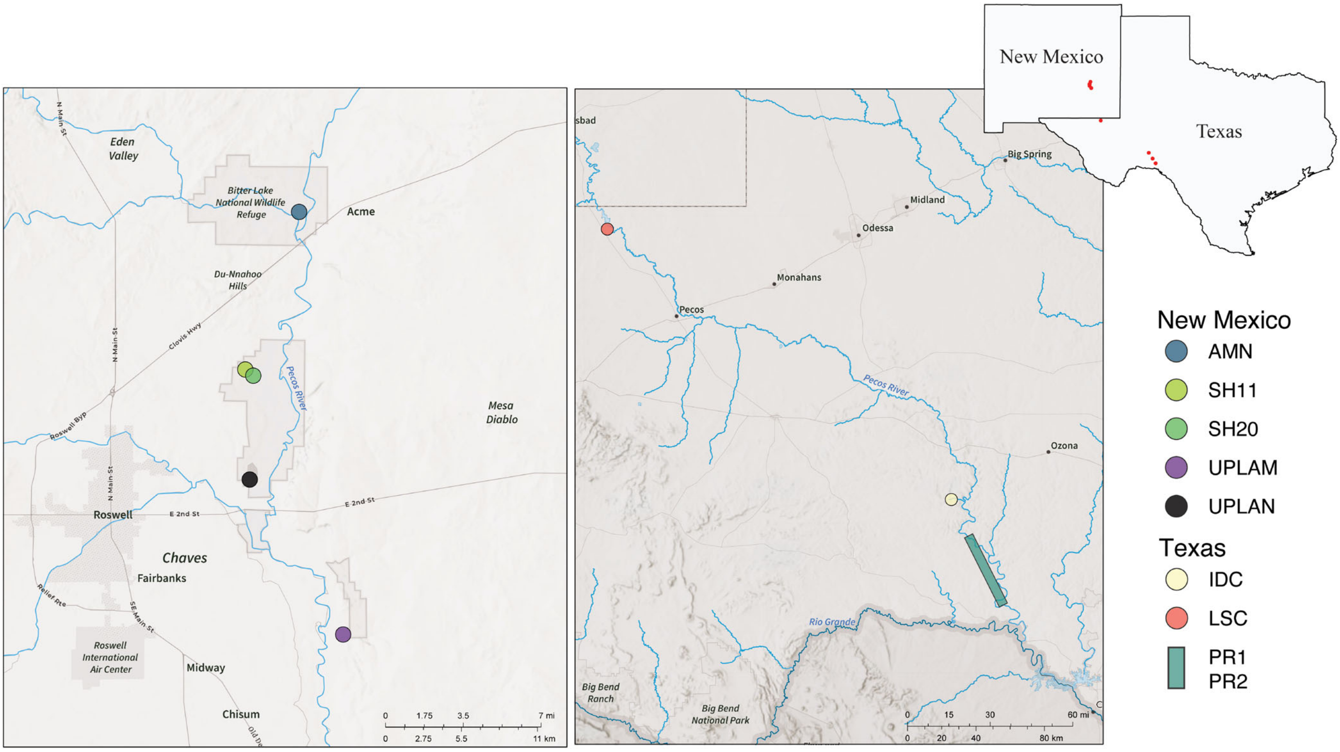

FIGURE 2 *Cyprinodon pecosensis* sampling locations in New Mexico and Texas. Two undisclosed locations within Texas are not shown: Upper Salt Creek (USC) and one reserve population of *C. pecosensis* location of a private land owner (PL). AMN, New Sinkhole, Chaves County, NM; SH11, Sinkhole 11, Chaves County, NM; SH20, Sinkhole 12, Chaves County, NM; UPLAM, Hunter Marsh Ditch, Chaves County, NM; UPLAN, BLM Overflow Wetland, Chaves County, NM; IDC, Independence Creek, Terrell County, TX; LSC, lower Salt Creek, Reeves County, TX; PR1 and PR2, private ranches, TX, locations obscured. Not shown: Fort Worth Zoo and the additional sampling locations for *C. variegatus*.

been well documented, broader population genetic studies are lacking, with little known about patterns of genetic diversity within and among the wild or refuge populations.

Due to the vulnerability of *C. pecosensis*, especially in Texas, assessing range-wide patterns of genetic variation is essential for scientifically guided population conservation and management. Therefore, the goal of this project was to use next generation sequencing techniques to conduct a conservation genomic assessment of hybridization and patterns of variation across more expansive sampling locations using thousands of genetic markers. The objectives of the research were to use next generation sequencing techniques to: (1) characterize contemporary and/or historical hybridization between *C. pecosensis* and *C. variegatus*; (2) assess standing genetic diversity within and among seemingly isolated groups of *C. pecosensis*; and (3) compare genetic diversity between wild populations of *C. pecosensis* and refuge populations.

2 | MATERIALS AND METHODS

2.1 | Sample collection

Fin clips were acquired across five locations in Texas, including two in Salt Creek and three areas along the mainstem of the Pecos River, and from five sites in New Mexico (Figure 2). These sites reflect both current and historical habitats for *C. pecosensis*. Fin clips were also obtained from two refuge populations seeded with individuals obtained from Salt Creek, including one housed on a private property in Pecos County, Texas (PL) and another housed at the Fort Worth Zoo, Fort Worth, Texas (FWZ). Additional sampling occurred elsewhere in Texas to characterize variation in *C. variegatus* (Galveston Seawall, Galveston, TX [GSW] and Tranquitas Creek, Kingsville, TX [TC]). Voucher specimens of *C. pecosensis*, *C. variegatus*, and their potential hybrids were also collected from multiple sites throughout the Pecos drainage in Texas. A complete list of sampling locations can be found in Table 1. All specimens have been deposited at the TAMU Biodiversity Research and Teaching Collections (TCWC), College Station, TX (Supporting Information Table S1).

2.2 | Genomic library preparation and sequencing

DNA was extracted using Mag-Bind Blood and Tissue DNA kits (Omega Bio-Tek, Norcross, GA) and approximately 1000 ng of high quality genomic DNA was used in a modified version of the double digest restriction-site associated DNA (ddRAD) library preparation method (Peterson et al., 2012). DNA extractions were digested with restriction enzymes *Eco*RI and *Msp*I and a barcoded adapter was

TABLE 1 *Cyprinodon pecosensis* sampling information including location, watershed, state (TX is Texas, NM is New Mexico).

Location	Watershed	State	Code	Lat	Long	*N*
Private Ranch 1	Pecos River	TX	PR1	–	–	10
Private Ranch 2	Pecos River	TX	PR2	–	–	3
Salt Creek	Salt Creek	TX	USC	–	–	37
Salt Creek	Salt Creek	TX	LSC	31.884	−103.922	14
TNC property	Independence Creek	TX	IDC	30.46	−101.806	10
BLM Overflow Wetland	Upper Pecos-Long Arroyo	NM	UPLAN	33.316	−104.342	36
Hunter Marsh Ditch	Upper Pecos-Long Arroyo	NM	UPLAM	33.417	−104.415	38
New Sinkhole	Arroyo del Macho	NM	AMN	33.591	−104.376	38
Sinkhole 11	Upper Pecos-Long Arroyo	NM	SH11	33.489	−104.418	40
Sinkhole 20	Upper Pecos-Long Arroyo	NM	SH20	33.485	−104.412	38
Private Lands	Pecos River	TX	PL*	–	–	39
Refuge	Fort Worth Zoo	TX	FWZ*	32.723	97.3564	40
Galveston Seawall	Galveston Bay	TX	GSW^	29.316	−94.761	7
Tranquitas Creek	Baffin Bay	TX	TC^	27.521	−97.84	2

Note: In two cases (PL and USC), latitude and longitude cannot be provided for legal reasons. Locations with reserve populations of *C. pecosensis* are denoted with an * and additional sites sampled for *C. variegatus* are denoted with a^.
Abbreviations: Code, location code; FWZ, fort worth zoo; IDC, independence creek; Lat, latitude; Long, longitude; LSC, location on Salt Creek; LSC, lower salt creek; *N*, number of individuals of *Cyprinodon* sequenced; PL, private lands; SH11, Sinkhole 11; SH20, Sinkhole 20; USC, upper Salt Creek; USC, upper salt creek.

ligated to *Eco*RI restriction sites while a common adapter was ligated to *Msp*I restriction sites. Individuals were subsequently pooled, using equimolar quantities of each ligated sample, into eight 'indexed' libraries consisting of 45 and 48 uniquely barcoded individuals per index and size selected using a Pippin Prep DNA size selection system (Sage Science Inc., Beverly, MA). Fragments were selected using a mean size of 375 bp, with a selection window of ±62 bp. Illumina flow-cell adapter sequences and index-specific identifiers were added to each index library, using 14 cycles of PCR. Each index was sequenced (paired end, 150 bp) as part of an Illumina NovaSeq 6000 sequence lane at Novogene (Sacramento, California).

2.3 | Data processing

Raw Illumina HiSeq reads from each individual were demultiplexed using the 'process_radtags' function in the software *Stacks* v2.59 (Catchen et al., 2013). Twenty four non-introgressed *C. pecosensis* individuals, selected from across sampling locations with no *C. variegatus* present (NM = 15, TX = 9), were used to create a de novo reference of putatively single-copy contigs, using the *dDocent* v2.9.1 pipeline (Puritz et al., 2014). All sequences were mapped to the reference and mapped reads were filtered to remove those which were not properly paired, those characterized as secondary alignments, and those which had a quality score below 40. Results were compiled into a variant call file (VCF) which identifies single nucleotide polymorphisms (SNPs) and variants were filtered using a combination of *VCFtools* v0.1.14 (Danecek et al., 2011) and custom R, BASH and Perl scripts to remove artefacts following O'Leary et al. (2018). These filters included removing loci with low average depth, loci with very high average depth, alleles which occur less than three times, genotypes with low quality scores, individuals with low average depth, genotypes with highly skewed allelic balance, loci influenced by library effects, monomorphic loci after removing individuals with high missing data, and loci with high depth variation across the locus. SNPs on the same fragment were then phased into microhaplotypes (hereafter loci) using *rad_haplotyper* (Willis et al., 2017) and loci which contained more haplotypes within individuals than expected were removed. Singletons and doubletons were filtered more stringently for depth than other alleles and duplicate individuals were removed. A complete description of bioinformatic processing can be found at https://github.com/marinegenomicslab/Pecos_pupfish_2025.

2.4 | Hybrid analysis

Principal component analysis (PCA) was performed in R using the package *adegenet* on the filtered SNP data to visualize the genetic relationship among all individuals (Supporting Information Figure S1; Jombart, 2008). Hybrids were assessed using Bayesian modelling of F1 and F1 cross categories in NewHybrids v2.0 with five independent runs (Anderson & Thompson, 2002). The reference individuals for *C. pecosensis* were from the refuge populations and the reference individuals for *C. variegatus* were collected from regions of Texas where only *C. variegatus* is known to reside. To test for various hybridization scenarios, simulated F1 hybrids, F2 hybrids and backcrossed individuals were created using the R package *adegenet* v2.1.10 (Jombart, 2008). Simulated individuals and sampled individuals were

TABLE 2 Pairwise estimates of F_{ST} (below the diagonal) and their corresponding *p*-values (above the diagonal) between *C. pecosensis* sampling locations.

	FWZ*	PL*	USC	AMN	SH11	SH20	UPLAM	UPLAN
FWZ*	–	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
PL*	0.0331	–	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
USC	0.0437	0.0298	–	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
AMN	0.2992	0.2997	0.2961	–	<0.0001	<0.0001	<0.0001	<0.0001
SH11	0.2995	0.3019	0.2984	0.2008	–	<0.0001	<0.0001	<0.0001
SH20	0.2573	0.2597	0.2574	0.1484	0.0905	–	<0.0001	<0.0001
UPLAM	0.2265	0.2287	0.2280	0.0926	0.1216	0.0667	–	<0.0001
UPLAN	0.3189	0.3193	0.3135	0.2193	0.2327	0.1863	0.1447	–

Note: The table has been partitioned to highlight comparisons within Texas and New Mexico (light grey) and between Texas and New Mexico (white). Locations with reserve populations of *C. pecosensis* are denoted with an * and additional sites sampled for *C. variegatus* are denoted with a^. Abbreviations: FWZ, Fort Worth Zoo; PL, Private Lands; SH11, Sinkhole 11; SH20, Sinkhole 20; USC, upper Salt Creek.

then visualized on a PCA with missing data imputed using empirically derived mean allele frequencies in the function scaleGen(), to see how sampled individuals grouped with simulated individuals. Additionally, genomic admixture was assessed using ADMIXTURE (Alexander et al., 2009), with $K = 2$ (corresponding to *C. pecosensis* and *C. variegatus*), to evaluate individual ancestry proportions and further elucidate patterns of hybridization. Simulation studies have shown reliable assignment of individuals back to parental species or hybrid status using an ancestry coefficient (q) threshold of 0.10 and 0.90 (Vähä & Primmer, 2006), however, individuals were assessed using more conservative cut offs ($q \geq 0.95$ or $q \leq 0.05$) based on recent fish hybridization literature (Brauer et al., 2023). To understand the directionality of hybridization or assess for historic introgression, mitochondrial DNA was sought from the ddRAD dataset by querying reference contigs against the National Center for Biotechnology Information nucleotide database using BLASTn (Camacho et al., 2009); however, no mitochondrial sequences were identified.

2.5 | Conservation genomics of *C. pecosensis*

Loci that are potentially under directional selection or balancing selection ('outlier loci') may provide information about selective pressures and adaptation at local and regional scales but can be misleading when the goal is to understand the roles that genetic drift and migration have played in genetic structuring of populations (Funk et al., 2012). Therefore, a dataset containing only non-introgressed *C. pecosensis* was screened for outliers using BAYESCAN (Foll & Gaggiotti, 2008) and OUTFLANK (Whitlock & Lotterhos, 2015) implemented in the R package *dartR* v1.8.3 (Gruber et al., 2018). BAYESCAN analysis was performed with 30 pilot runs of 5000 iterations and a burn-in of 50,000 iterations before sampling 5000 times at 100 iteration intervals, a prior odds value of 100 and an q-value of 0.05. In OUTFLANK, the F_{ST} distribution was made by trimming the upper and lower 5% and the q-value was set to 0.1. Datasets were tested when grouped by location, but due to large genetic differences between the states (Table 2), locations within each state were tested by themselves as well.

Homogeneity of genetic variation was assessed using a single level, locus-by-locus analysis of molecular variation (AMOVA) implemented in ARLEQUIN v3.5 (Excoffier & Lischer, 2010), which accommodates uneven levels of missing data across loci (Weir & Cockerham, 1984). Significance was assessed at an α-level of 0.05 using 10,000 permutations in ARLEQUIN v3.5. Post-hoc pairwise F_{ST} was estimated in ARLEQUIN with significance determined as above, but correcting for multiple testing using the Benjamini and Hochberg (1995) false discovery rate (BHFDR) procedure. Finally, an unsupervised clustering algorithm (K-means) was used to group the individuals, with the optimal number of groups picked using Bayesian information criterion (BIC). These groups were then tested using discriminant analysis of principal components (DAPC) with cross validation, in the R package *adegenet* (Jombart, 2008).

For each location expected heterozygosity (H_e) was estimated using ARLEQUIN and rarified allelic richness (A_R) was estimated using *hierfstat* (Goudet & Jombart, 2022). Within-group diversity measures were compared among groups using a Friedman's rank sum test in the R package *stats* v 3.6.0 (R Core Team, 2019), with post hoc pairwise comparisons made using a paired Wilcoxon's signed-rank test, performed in the R package *coin* 1.4-1 (Hothorn et al., 2008) which adjusts for tied rank values. Significance values were corrected for multiple comparisons using BHFDR. Contemporary effective population size (N_E) was estimated for each location using the linkage disequilibrium approach implemented in NEESTIMATOR v2.1 (Do et al., 2014;) with a correction for physical linkage (Waples et al., 2016).

2.6 | Ethics statement

All voucher specimens and tissue samples (caudal fin clips) were acquired following appropriate animal care standards of the individual Federal and State agencies involved.

3 | RESULTS

Fin clips from 354 *Cyprinodon* were collected and sequenced across eight ddRAD indices resulting in 2,498,928 SNPs across 103,611 putative loci. Final assembly parameters for the de novo reference were a *c*-value of 0.88, a K_1 of 2 and a K_2 of 1. After filtering, 18,085 SNPs across 5156 SNP-containing loci were present in 352 individuals, with an average of 3.5 SNPs and 3.7 alleles per locus.

3.1 | Hybrid analysis

No individuals from New Mexico or the Texas refuge populations were found to be hybrids using NewHybirds. Within Texas, five sites were sampled in the wild and some history of hybridization was found in all of them, except Upper Salt Creek (USC). In USC, at an undisclosed location, both analyses indicated that there were only non-introgressed *C. pecosensis* individuals ($n = 37$; Supporting Information Figure S2). At a second location on Salt Creek (LSC), closer to the confluence with the Pecos, NewHybirds indicated half of the individuals were *C. variegatus* backcrosses, two were *C. variegatus*, and additional five individuals were undetermined. The undetermined individuals were either identified as *C. variegatus* or *C. variegatus* backcrosses across multiple runs of NewHybirds. Results of PCA supported these conclusions, showing that all individuals were admixed with a tendency toward backcrossing with *C. variegatus* (Supporting Information Figure S3). In Independence Creek (IDC, $n = 10$) nine *C. variegatus* were found as well as one individual categorized by NewHybirds as a *C. variegatus* backcross. In the two sites, sampled along the Pecos River (PR1; $n = 10$, PR2 $n = 3$) NewHybirds identified one *C. variegatus* backcross and could not determine if one individual was non-introgressed *C. variegatus* or a *C. variegatus* backcross, all other individuals were identified as *C. variegatus*. Results of the PCA supported these conclusions, showing most individuals to be very close to non-introgressed *C. variegatus* (Supporting Information Figure S4). Results of admixture analysis were consistent with the results NewHybirds and hybrid simulations (Figure 3). All *C. pecosensis* individuals from New Mexico were categorized as pure, $q < 0.05$. In all Texas wild populations, except USC, results indicated high levels of admixture between *C. pecosensis* and *C. variegatus*. In USC several individual *C. pecosensis* had ancestry coefficients above the 0.05 threshold, ranging from $q = 0.064$ to $q = 0.097$.

3.2 | Conservation genomics of *C. pecosensis*

After removing *C. variegatus* and admixed individuals, 306 individuals remained with 18,056 SNPs across 5148 SNP-containing loci. OUTFLANK did not return any outlier loci putatively under selection; however, 89 outlier loci were identified using BAYESCAN among all the sites ($n = 8$) and within NM ($n = 5$). These loci were removed from the dataset resulting in a final neutral dataset that contained 306 individuals, genotyped at 17,725 SNPs across 5059 loci, with an average of 3.5 SNPs and 2.9 alleles per locus.

Significant genetic heterogeneity was found among locations (% $V = 21.63$, F_{ST}: 0.21635, $p < 0.0001$; Supporting Information Table S2). Pairwise estimates of F_{ST} were significant between all locations after correcting for multiple comparisons ($p < 0.001$; Table 2). Estimates between USC and New Mexico sites were relatively large (0.228–0.314), while the estimates between New Mexico locations were mostly smaller (0.067–0.233). The refuge populations (FWZ and PL) were more similar to USC and each other (F_{ST}, 0.030–0.044) than they were to any New Mexico sites (F_{ST}, 0.229–0.319). The minimum BIC value was at $K = 6$; however, the rate of change in BIC decreased greatly after $K = 4$, suggesting this could represent higher order structuring (Supporting Information Figure S5). Both values of *K* had 100% reassignment in cross validation. A *K*-means clustering of four groups placed the Texas individuals, including the refuge population, together in one group while both sinkholes north of Bitter Lake (Sinkhole 11; SH11, Sinkhole 20; SH20) also grouped together. The sample 12 km north of Bitter Lake (New Sinkhole; AMN), along the Pecos River grouped together with the sample on the south side of Bitter Lake (Hunter Marsh Ditch; UPLAM). The sample 13 km south of Bitter Lake near Lea Lake (Bureau of Land Management Overflow Wetleand; UPLAN) was identified as a separate group. When six groups were considered, all sampling locations in New Mexico were found to be separate while the Texas sampling locations still grouped together (Figure 4).

There were significant differences among locations in H_E ($X^2 = 906.3$; $p < 0.001$) and A_R ($X^2 = 1520.3$; $p < 0.001$). Expected heterozygosity ranged from 0.121 (Refuge; PL) to 0.157 (UPLAM) and the pairwise comparisons were significant in all but two comparisons (SH11 and UPLAN; SH20 and USC). Allelic richness ranged from 1.562 (SH11) to 1.836 (UPLAM) and pairwise comparisons were significant for 24 out of 28 comparisons (AMN and UPLAN; FWZ and SH11; PL and UPLAN; UPLAM and USC). The refuge population (FWZ) had the lowest point estimate of effective population size (12.0) and USC and PL were both under 50. By contrast, all of New Mexico locations had estimates N_e over 500, with the sample at the south end of Bitter Lake (UPLAM) estimated to have an N_e of over 8300 (Table 3).

4 | DISCUSSION

Introductions of non-native congeners can pose significant threats to imperilled species of freshwater fishes due to competition, hybridization and genetic swamping (Allendorf, 1991; Perry et al., 2002). Documentation of hybridization and genetic swamping is prevalent in important game fishes in North America including various species of trout and bass (Allendorf & Leary, 1988; Bangs et al., 2018; Dowling & Childs, 1992; Krueger & May, 1991; McCracken et al., 1993; Philipp et al., 1983; Smith et al., 2025; Whitmore, 1983). Understanding the consequences of genetic introgression on threatened, non-game freshwater fishes is less robust, with the most notable examples involving chubs in the genera *Gila* and *Siphateles*, sticklebacks in the genus *Gasterosteus*, and the widespread

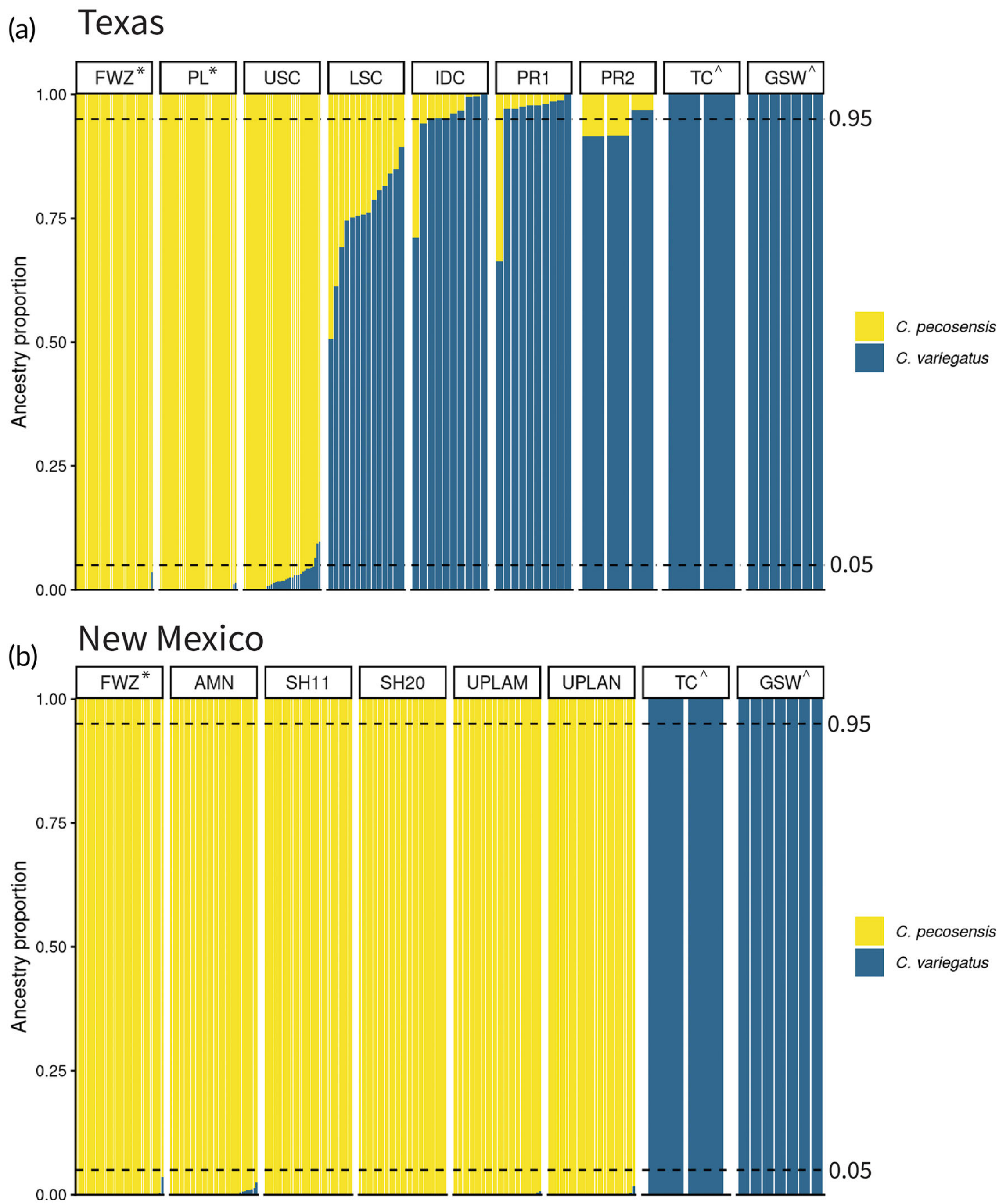

FIGURE 3 Ancestry plots from the ADMIXTURE analysis between *C. pecosensis* and *C. variegatus* in Texas (a) and New Mexico (b). Dashed lines indicate ancestry coefficients of $q = 0.05$ or 0.95. Locations with reserve populations of *C. pecosensis* are denoted with an * and additional sites sampled for *C. variegatus* are denoted with a ^.

introduction of *C. variegatus* and its effects on other species of *Cyprinodon* (Ayers, 2018; Black et al., 2017; Chen et al., 2013; Echelle et al., 1997; Moyle, 1976). *C. pecosensis* in the state of Texas, is threatened with extirpation, in large part because of interactions (i.e., competition and hybridization) with the introduced congener *C. variegatus* (Childs et al., 1996). Of the five sites sampled in Texas (two on the Pecos River mainstem, two on Salt Creek and one on Independence Creek), non-introgressed *C. pecosensis* individuals were found only at a single site on Upper Salt Creek (USC) that is spatially isolated from the confluence by a stretch of dry riverbed. Although three individuals in USC have ancestry coefficients above 0.05 ($q = 0.064–0.097$), these levels are much lower than expected from F1 ($q \approx 0.50$) or backcrossed ($q \approx 0.25$) hybrids but could be indicative of historic contact between the species, despite there being no record of *C. variegatus* being present in USC (M. Bean personal communication). Given that *C. pecosensis* and *C. variegatus* are closely related,

JOURNAL OF FISH BIOLOGY fsbi

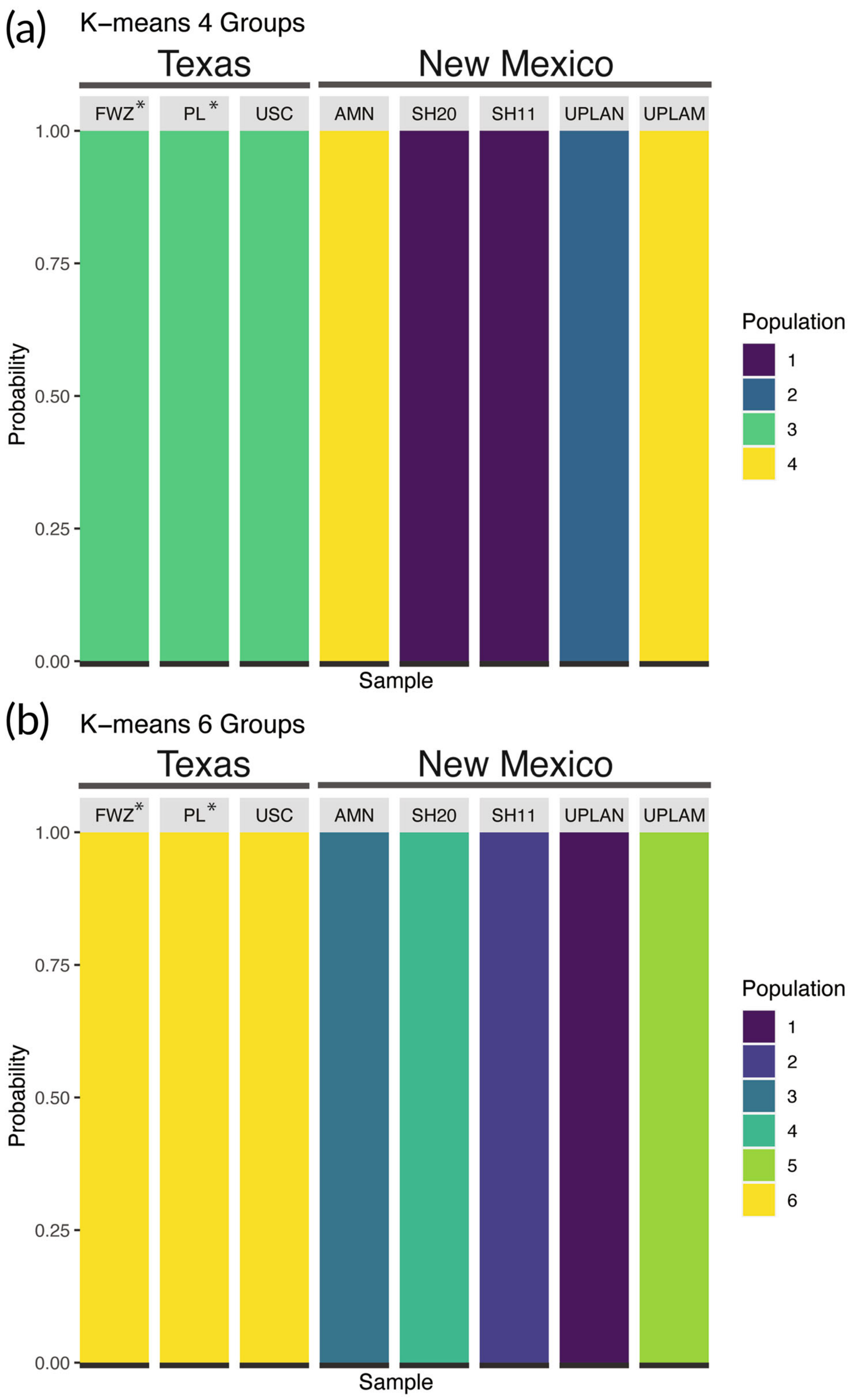

FIGURE 4 Discriminant analysis of principal components (DAPC) cross validation. (a) results with a *k*-means of four (b) results with a *k*-means of six. For each column at each location, the solid bar indicates 100% reassignment of the individual to the group. Locations with reserve populations of *C. pecosensis* are denoted with an *.

ancestry coefficients below 0.10 may also reflect shared ancestral polymorphisms or statistical noise (Lawson et al., 2018; Maddison, 1997), rather than admixture. At all other sites individuals were either admixed or *C. variegatus*, with the results supporting the notion that *C. variegatus* is genetically swamping out *C. pecosensis* in most of its former range in Texas (Garrett et al., 2002). The two refuge populations (FWZ and PL) contained only non-introgressed *C. pecosensis*, consistent with the fact that they were originally sourced from USC where no interaction between the two species is currently occurring. While this study has only confirmed one predominantly non-introgressed, wild population of *C. pecosensis* in Texas, the species has been documented to occur on private property elsewhere

TABLE 3 Estimates of genetic diversity for *C. pecosensis*, including expected Heterozygosity (H_e), allelic richness (A_r) and effective size (N_E) by catch location.

					N_E		
Location	State	n	H_e	A_r	Lower	Point	Upper
FWZ*	TX	40	0.122	1.563	8.7	12.0	16.8
PL*	TX	39	0.1218	1.604	21.1	31.7	53.6
USC	TX	37	0.1352	1.819	29.4	41.7	66.0
AMN	NM	38	0.1422	1.58	283.5	592.5	Infinite
SH11	NM	40	0.1374	1.562	1016.3	1869.7	11,194.5
SH20	NM	38	0.1512	1.693	532.1	792.3	1536.1
UPLAM	NM	38	0.1567	1.836	1956.4	8344.0	Infinite
UPLAN	NM	36	0.1397	1.592	742.5	1065.3	1874.7

Note: Effective size (*point*) estimates using the linkage disequilibrium approach and corrected for physical linkage for each location. Estimates used a minor allele frequency cutoff of 0.02 and included a 95% confidence interval (lower, upper) estimated parametrically. Locations with reserve populations of *C. pecosensis* are denoted with an * and additional sites sampled for *C. variegatus* are denoted with a ^. Abbreviations: SH11, Sinkhole 11; SH20, Sinkhole 20; USC, upper Salt Creek.

on Salt Creek past the dry section (Echelle & Echelle, 1978). If still present, it seems highly unlikely that this population has been impacted by *C. variegatus*. By contrast, no *C. variegatus* were collected in New Mexico and no evidence of admixture was detected in the New Mexico individuals of *C. pecosensis*.

While efforts to cull *C. variegatus* seem to have been successful for maintaining the genetic integrity of other species of *Cyprinodon* (e.g., *C. bovinus*; Black et al., 2017), efforts to remove the invasive congener over the entirety of their introduced range within the Pecos basin may prove too difficult to be effective for *C. pecosensis*. For example, large scale removal efforts to eradicate western mosquito fish (*Gambusia affinis* Baird & Girard, 1853) and eastern mosquito fish (*G. holbrooki* Girard, 1859b) have been attempted using poisoning or electroshocking methods with little success (Kalogianni et al., 2025; Pyke, 2008). The former method results in unintentional damage to other aquatic fauna, while the latter, even though it is more targeted, seems to only be effective as a population control measure. Additionally, hybrid swarms between *C. pecosensis* and *C. variegatus* develop rapidly, perhaps because admixed individuals exhibit reproductive advantages; female *C. pecosensis* preferentially mate with male *C. variegatus*, and both *C. variegatus* and hybrid males have been observed to be more aggressive, leading to higher mating success (Kodric-Brown & Rosenfield, 2004; Rosenfield et al., 2004; Rosenfield & Kodric-Brown, 2003).

Assessment of non-introgressed populations of *C. pecosensis*, resulted in comparisons of three Texas locations (FWZ, PL and USC) and five New Mexico locations (AMN, SH11, SH20, UPLAM, UPLAN). Across analyses there was a clear distinction between *C. pecosensis* populations in New Mexico and Texas, indicating deep genetic divergence. Historically, *C. pecosensis* was widespread throughout the Pecos River drainage but has seen population declines and large range contractions (Echelle & Echelle, 2003; Hoagstrom et al., 2025). Events such as the construction of Red Bluff Dam, just north of Pecos, Texas in the 1930's (Jensen et al., 2006) cut off northern reaches of the Pecos river in New Mexico from Texas. Dams on the river, degraded habitat, and increased water usage resulting in dry stretches of the Pecos River and its tributaries, may serve as barriers to gene flow, allowing strong genetic drift to act on small, isolated populations (Coleman et al., 2018; Hoagstrom & Brooks, 1999; Miller, 1961; Roberts et al., 2013).

In Texas, significant differentiation was detected between USC, FWZ and PL despite the recent establishment of the refuge populations using individuals collected from USC (FWZ in 2012, with supplemental introductions in 2015 and 2018, and PL in 2014; personal communication M. Bean and M. Montagne). However, estimates of N_e for all three populations were small (USC, 29; PL, 21; FWZ, 12) and generation times for pupfish are likely a year or less (Garrett et al., 2002), suggesting that accelerated drift likely explains the pattern (Wright, 1931). Upper Salt Creek had elevated heterozygosity and allelic richness relative to PL and FWZ, which could be due to the fact that USC was the source used to create both refuge populations, but it could also indicate intermittent connectivity between USC and other locations of *C. pecosensis* in Salt Creek. Genetic differentiation is common in other *Cyprinodon* species which have associated refugia populations. In both *C. bovinus* and *C. tularosa* Miller and Echelle 1975 (White Sands pupfish), there are moderate levels of divergence between captive and wild populations which has been attributed to genetic drift due to founder effects, variability of reproductive success and population instability in the wild (Black et al., 2017, 2024; Bretzing-Tungate et al., 2026; Wilcox & Martin, 2006). It is important to note that contemporary estimates of N_e at or below 50 are taken to indicate that populations are at risk of rapidly losing genetic diversity, which increases the probability of extirpation and extinction (Frankham et al., 2014). In such scenarios, intervention via genetic rescue (i.e., increasing the populations fitness via gene flow; Hedrick et al., 2011) may be warranted; however, identifying a suitable source population poses a challenge, particularly given the reduced diversity at the refuge populations. Additional locations on Salt Creek may represent a reservoir of genetic diversity in Texas and could serve as an additional source for assisted gene flow (i.e., translocation of individuals among populations to increase genetic diversity; Pavlova et al., 2017), but it remains uncertain if *C. pecosensis* is still present at those sites.

In New Mexico, all populations were significantly diverged from one another but were found to have high diversity relative to the remaining Texas population with no evidence of hybridization. Once more widespread in New Mexico, *C. pecosensis* currently occupies a fragmented distribution across isolated sinkholes, man-made lakes, main stem habitat and spring-fed marshes (Collyer et al., 2015; Hoagstrom & Brooks, 1999). While sampling in New Mexico was not exhaustive, it included populations in both sinkholes (AMN, SH11 and SH20) and marsh habitats (UPLAM and UPLAN). Of all sampled New Mexico populations, UPLAN appeared most genetically distinct (mean $F_{ST} = 0.248$), likely reflecting long-term isolation. This sample was taken in a marsh habitat which is isolated from the other sites in this study and only occasionally connected to the Pecos mainstem during high flow events, though a physical barrier at the outlet may limit upstream movement making any gene flow unidirectional. The second most diverged site, AMN, is a sinkhole location, which is also distant from other locations, but closer to the Pecos River than UPLAN. The estimated N_E of AMN (592) is about half that of UPLAN and smaller than any other sampled population in New Mexico, potentially due to smaller, less suitable habitat (Martin et al., 2016). The small effective size of this population may explain its relatively large divergence from all other sites, as effects of genetic drift are more pronounced in populations with small effective sizes (Wright, 1931). The remaining three sites are closer in proximity and exhibit pairwise genetic divergences of relatively low magnitude. These sites include UPLAM, another large marsh system, that has physical connectivity to the nearby marshes and streams, likely to explain increased genetic diversity and N_E observed here. Finally, the effective size for SH11 is much larger than all other sinkhole locations, although the population exhibits lower allelic richness and heterozygosity, suggesting that SH11 may represent a demographically stable, low-disturbance habitat that may have reduced diversity due to a founder event (Nei et al., 1975).

The overarching pattern of genetic variation across New Mexico populations is likely based on a combination of the amount and suitability of habitats (i.e. marshes vs. sinkholes), location (i.e. proximity to the Pecos River and other populations) as well as historical demography. Historically, sinkholes, marshes and wetlands experienced connectivity via cycles of flooding from the Pecos mainstem, followed by periods of drying, isolating habitats (Hoagstrom & Brooks, 1999). Since the late 1800s, there has been increased water usage for anthropogenic activities, water diversion for agriculture, and river damming, all contributing to a decreases in the frequency and strength of flood events and an increase in isolation of populations of *C. pecosensis* (Collyer et al., 2015; Hoagstrom & Brooks, 1999). This is reflected in the data, which shows significant differences between all New Mexico locations. Even locations which are physically close to one another (SH11 and SH20; <1 km straight distance) were significantly diverged, with significantly different estimates of within population diversity. While some differences were relatively small in magnitude, levels of divergence were consistent with that reported for other pupfishes (Finger et al., 2013). Much of the habitat currently occupied by *C. pecosensis* in New Mexico formed from slowly dissolving gypsum deposits followed by ground water penetration. Contemporary connectivity among *C. pecosensis* populations in New Mexico likely depends on monsoon rains that vary spatially and temporally (Adams and Comrie, 1997) and subsequent flood events, which can temporarily link nearby habitats, allowing for movement of individuals and gene flow (Hoagstrom & Brooks, 1999; Kalra et al., 2021). Given reductions in contemporary flooding events and lack of evidence for subterranean connections (Hoagstrom & Brooks, 1999; Johnson, 1996), many of these sites may be intermittently or permanently isolated. Contrary to findings in Texas, sites in New Mexico, which were diverged from one another had estimates of N_e in the thousands, suggest longer periods of isolation but also that near term loss of genetic diversity is less of a concern. A more comprehensive study of *C. pecosensis*, spatially and temporally, will be needed to disentangle the effects of flooding, habitat size and habitat quality, which can have implications for species persistence.

Taken as a whole, the results of this study indicate that *C. pecosensis* in Texas is in a perilous position. The geographically widespread genetic swamping found here, along with studies showing potential hybrid vigour for *C. pecosensis*/*C. variegatus* crosses (Rosenfield et al., 2004), indicates that all possible precautions should continue to prevent connections from being established between lower Salt Creek and the upper stretch where non-introgressed *C. pecosensis* persist (potentially the last remaining wild population of this species in the state). However, low levels of standing diversity indicate that the wild population in USC is primed to enter the extinction vortex (Gilpin & Soulé, 1986). One potential solution would be assisted gene flow to increase overall genetic diversity (Pavlova et al., 2017). While the refuge populations and USC were genetically distinct, those differences were of a relatively small magnitude, so while it is unlikely that assisted gene flow would have a negative effect, it is also unclear whether there would be large benefits. The New Mexico sites on the other hand are more genetically diverse, providing a larger pool of variation that could be introduced to USC. However, estimated levels of divergence between New Mexico populations and USC were quite large (F_{ST}, 0.23–0.32), suggesting the potential for genetic incompatibilities, maladaptive variation and/or differences in mating preference (Hedrick et al., 2019; Twardek et al., 2023) that might result in unexpected negative impacts if assisted gene flow was attempted. One solution to this dilemma would be the creation of an experimental captive population, seeded with genotyped individuals from New Mexico and USC, followed by rigorous genetic monitoring over the course of several generations to ensure that mixing the two gene pools is a viable option.

AUTHOR CONTRIBUTIONS

Project conception: Megan G. Bean, David S. Portnoy and Kevin W. Conway. *Data generation*: Megan G. Bean, Kevin W. Conway, Elizabeth P. Dolan, Robyn R. Doege, Andrew T. Fields, Joanna L. Hatt and David S. Portnoy. *Data analysis*: Elizabeth P. Dolan, Andrew T. Fields and David S. Portnoy. *Manuscript preparation—original draft*: Elizabeth P. Dolan. *Manuscript preparation—review and editing*: Megan G. Bean, Kevin W. Conway, Robyn R. Doege, Andrew T. Fields, Joanna L. Hatt and David S. Portnoy. *Funding*: David S. Portnoy, Kevin W. Conway, Megan G. Bean and Joanna L. Hatt.

ACKNOWLEDGEMENTS

This is publication 48 of the Marine Genomics Laboratory and 140 of Genetic Studies in Fishes, and publication 1728 of the Biodiversity Research and Teaching Collections at Texas A&M University. We are grateful to the Texas private landowners who provided access to the Pecos River through their properties. We especially thank the landowner who collaborated with state and federal partners to establish a secure refuge population for *C. pecosensis* and who remains dedicated to the conservation of this species in Texas. We would like to thank Alice Best of the Inland Fisheries division at Texas Parks and Wildlife Department for assistance in this project. We also thank Nathan Allan and Gary Garrett for sharing information on historical collections. Fieldwork support was generously provided by Michael Montagne, Maritza Mallek, Carl Jacobsen, JC Boothe, Bryan Bakevich, Nathan Thompson and the staff at Bitter Lake National Wildlife Refuge. The findings and conclusions in this article are those of the author(s) and do not represent the official views of the US Fish and Wildlife Service.

FUNDING INFORMATION

Research was funded by Texas Parks and Wildlife Department, State Wildlife Grant Program (F21AP03616).

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Elizabeth P. Dolan https://orcid.org/0000-0002-3443-5323
Andrew T. Fields https://orcid.org/0000-0002-2817-2141
David S. Portnoy https://orcid.org/0000-0002-8178-1018

REFERENCES

Adams, D. K., & Comrie, A. C. (1997). The North American Monsoon. *Bulletin of the American Meteorological Society*, *78*, 2197–2213.

Alexander, D. H., Novembre, J., & Lange, K. (2009). Fast model-based estimation of ancestry in unrelated individuals. *Genome Research*, *19*(9), 1655–1664. https://doi.org/10.1101/gr.094052.109

Allendorf, F. W. (1991). Ecological and genetic effects of fish introductions: Synthesis and recommendations. *Canadian Journal of Fisheries and Aquatic Sciences*, *48*(S1), 178–181. https://doi.org/10.1139/f91-318

Allendorf, F. W., & Leary, R. F. (1988). Conservation and distribution of genetic variation in a polytypic species, the cutthroat trout. *Conservation Biology*, *2*(2), 170–184. https://doi.org/10.1111/j.1523-1739.1988.tb00168.x

Anderson, E. C., & Thompson, E. A. (2002). A model-based method for identifying species hybrids using multilocus genetic data. *Genetics*, *160*(3), 1217–1229. https://doi.org/10.1093/genetics/160.3.1217

Ayers, K. (2018). *Analysis of hybridization between native and invasive pupfishes (Cyprinodon) (Masters)* (p. 87). Stephen F. Austin State University, Nacogdoches.

Baird, S. F., & Girard, C. F. (1853). Descriptions of new species of fishes collected by Mr. John H. Clark, on the U. S. and Mexican boundary survey, under Lt. Col. Jas. D. Graham. In *Proceedings of the Academy of Natural Sciences of Philadelphia* (Vol. 6, pp. 387–390). Academy of Natural Sciences.

Bangs, M. R., Oswald, K. J., Greig, T. W., Leitner, J. K., Rankin, D. M., & Quattro, J. M. (2018). Introgressive hybridization and species turnover in reservoirs: A case study involving endemic and invasive basses (Centrarchidae: *Micropterus*) in southeastern North America. *Conservation Genetics*, *19*(1), 57–69. https://doi.org/10.1007/s10592-017-1018-7

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B, Statistical Methodology*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Black, A. N., Heenkenda, E. J., Mathur, S., Willoughby, J. R., Pierce, B. L., Turner, S. J., Rizzuto, D., & DeWoody, J. A. (2024). Rapid vertebrate speciation via isolation, bottlenecks, and drift. *Proceedings of the National Academy of Sciences*, *121*(22), e2320040121. https://doi.org/10.1073/pnas.2320040121

Black, A. N., Seears, H. A., Hollenbeck, C. M., & Samollow, P. B. (2017). Rapid genetic and morphologic divergence between captive and wild populations of the endangered Leon Springs pupfish, Cyprinodon bovinus. *Molecular Ecology*, *26*(8), 2237–2256. https://doi.org/10.1111/mec.14028

Blomqvist, D., Pauliny, A., Larsson, M., & Flodin, L.-Å. (2010). Trapped in the extinction vortex? Strong genetic effects in a declining vertebrate population. *BMC Evolutionary Biology*, *10*(1), 33. https://doi.org/10.1186/1471-2148-10-33

Brauer, C. J., Sandoval-Castillo, J., Gates, K., Hammer, M. P., Unmack, P. J., Bernatchez, L., & Beheregaray, L. B. (2023). Natural hybridization reduces vulnerability to climate change. *Nature Climate Change*, *13*(3), 282–289. https://doi.org/10.1038/s41558-022-01585-1

Bretzing-Tungate, R. J., Conway, K. W., Bean, M. G., Smith, R. K., Hollenbeck, C. M., Fields, A. T., Dolan, E. P., Mussmann, S. M., & Portnoy, D. S. (2026). Conservation genomics of the imperiled Leon springs pupfish, *Cyprinodon bovinus*, across temporal and spatial scales. *Conservation Genetics*, *27*, 8. https://doi.org/10.1007/s10592-025-01743-w

Burkhead, N. M. (2012). Extinction rates in north American freshwater fishes, 1900–2010. *Bioscience*, *62*(9), 798–808. https://doi.org/10.1525/bio.2012.62.9.5

Calamusso, B., Rinne, J. N., & Edwards, R. J. (2005). *Historic changes in the Rio Grande fish fauna: status, threats, and management of native species* (Vol. 45, pp. 205–223). American Fisheries Society Symposium.

Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K., & Madden, T. L. (2009). BLAST+: Architecture and applications. *BMC Bioinformatics*, *10*(1), 421. https://doi.org/10.1186/1471-2105-10-421

Catchen, J., Hohoenlohe, P. A., Bassham, S., Amores, A., & Cresko, W. (2013). Stacks an analysis tool set for population genomics. *Molecular Ecology*, *22*(11), 3124–3140.

Cheek, C. A., & Taylor, C. M. (2016). Salinity and geomorphology drive long-term changes to local and regional fish assemblage attributes in the lower Pecos River, Texas. *Ecology of Freshwater Fish*, *25*(3), 340–351. https://doi.org/10.1111/eff.12214

Chen, Y., Parmenter, S., & May, B. (2013). Genetic characterization and management of the endangered Mohave tui chub. *Conservation Genetics*, *14*(1), 11–20. https://doi.org/10.1007/s10592-012-0410-6

Childs, M. R., Echelle, A. A., & Dowling, T. E. (1996). Development of the hybrid swarm between Pecos pupfish (Cyprinodontidae: *Cyprinodon pecosensis*) and sheepsheads minnow (*Cyprinodon variegatus*): A perspective from allozymes and mtDNA. *Evolution*, *50*(5), 2014–2022. https://doi.org/10.1111/j.1558-5646.1996.tb03588.x

Coleman, R. A., Gauffre, B., Pavlova, A., Beheregaray, L. B., Kearns, J., Lyon, J., Sasaki, M., Leblois, R., Sgro, C., & Sunnucks, P. (2018). Artificial barriers prevent genetic recovery of small isolated populations of a low-mobility freshwater fish. *Heredity*, *120*(6), 515–532. https://doi.org/10.1038/s41437-017-0008-3

Collyer, M. L., Hall, M. E., Smith, M. D., & Hoagstrom, C. W. (2015). Habitat-morphotype associations of Pecos pupfish (*Cyprinodon pecosensis*) in isolated habitat complexes. *Copeia*, *103*(1), 181–199. https://doi.org/10.1643/ot-14-084

Craig, C. A., & Bonner, T. H. (2019). Drainage basin checklists and dichotomous keys for inland fishes of Texas. *ZooKeys*, *874*, 31–45. https://doi.org/10.3897/zookeys.874.35618

Danecek, P., Auton, A., Abecasis, G., Albers, C. A., Banks, E., DePristo, M. A., Handsaker, R. E., Lunter, G., Marth, G. T., Sherry, S. T., McVean, G., Durbin, R., & 1000 Genomes Project Analysis Group. (2011). The variant call format and VCFtools. *Bioinformatics*, *27*(15), 2156–2158. https://doi.org/10.1093/bioinformatics/btr330

Do, C., Waples, R. S., Peel, D., Macbeth, G. M., Tillett, B. J., & Ovenden, J. R. (2014). NeEstimator v2: Re-implementation of software for the estimation of contemporary effective population size (N_e) from genetic data. *Molecular Ecology Resources*, *14*(1), 209–214. https://doi.org/10.1111/1755-0998.12157

Dowling, T. E., & Childs, M. R. (1992). Impact of hybridization on a threatened trout of the southwestern United States. *Conservation Biology*, *6*(3), 355–364. https://doi.org/10.1046/j.1523-1739.1992.06030355.x

Echelle, A. A., & Connor, P. J. (1989). Rapid, geographically extensive genetic introgression after secondary contact between two pupfish species (*Cyprinodon*, Cyprinodontidae). *Evolution*, *43*(4), 717–727. https://doi.org/10.1111/j.1558-5646.1989.tb05171.x

Echelle, A. A., & Echelle, A. F. (1978). The Pecos River pupfish, *Cyprinodon pecosensis* n. Sp. (Cyprinodontidae), with comments on its evolutionary origin. *Copeia*, *1978*(4), 569. https://doi.org/10.2307/1443683

Echelle, A. A., & Echelle, A. F. (2020). Cyprinodontidae: Pupfishes. In M. L. W. Jr & B. M. Burr (Eds.), *Freshwater fishes of North America: Characidae to Poecilidae* (Vol. 2, pp. 609–673). John Hopkins University Press.

Echelle, A. A., Echelle, A. F., Balderas, S. C., & Vilano, M. L. (2003). Pupfishes of the northern Chihuahuan desert: Status and conservation. In *Aquatic Fauna of the northern Chihuahuan Desert* (Vol. 46, pp. 113–126). Museum of Texas Tech University, Special Publications.

Echelle, A. A., Echelle, A. F., & Edds, D. R. (1987). Population structure of four pupfish species (Cyprinodontidae: *Cyprinodon*) from the Chihuahuan Desert region of New Mexico and Texas: Allozymic variation. *Copeia*, *1987*(3), 668. https://doi.org/10.2307/1445659

Echelle, A. A., Hoagstrom, C. W., Echelle, A. F., & Brooks, J. E. (1997). Expanded occurrence of genetically Introgressed pupfish (Cyprinodontidae: Cyprinodon pecosensis X variegatus) in New Mexico. *The Southwestern Naturalist*, *42*(3), 336–339.

Echelle, A. A., & Miller, R. R. (1974). Rediscovery and Redescription of the Leon Springs pupfish, Cyprinodon bovinus, from Pecos County, Texas. *The Southwestern Naturalist*, *19*(2), 179. https://doi.org/10.2307/3670278

Edds, D. R., & Echelle, A. A. (1989). Genetic comparisons of hatchery and natural stocks of small endangered fishes: Leon Springs pupfish, Comanche Springs pupfish, and Pecos gambusia. *Transactions of the American Fisheries Society*, *118*, 441446.

Elkins, L. C., Acre, M. R., Bean, M. G., Robertson, S. M., Smith, R., & Perkin, J. S. (2024). A multiscale perspective for improving conservation of Conchos pupfish. *Animal Conservation*, *27*(4), 538–553. https://doi.org/10.1111/acv.12930

Excoffier, L., & Lischer, H. E. L. (2010). Arlequin suite ver 3.5: A new series of programs to perform population genetics analyses under Linux and windows. *Molecular Ecology Resources*, *10*(3), 564–567. https://doi.org/10.1111/j.1755-0998.2010.02847.x

Finger, A. J., Parmenter, S., & May, B. P. (2013). Conservation of the Owens pupfish: Genetic effects of multiple translocations and extirpations. *Transactions of the American Fisheries Society*, *142*, 1430–1443.

Foll, M., & Gaggiotti, O. (2008). A genome-scan method to identify selected loci appropriate for both dominant and codominant markers: A Bayesian perspective. *Genetics*, *180*(2), 977–993. https://doi.org/10.1534/genetics.108.092221

Frankham, R., Bradshaw, C. J. A., & Brook, B. W. (2014). Genetics in conservation management: Revised recommendations for the 50/500 rules, red list criteria and population viability analyses. *Biological Conservation*, *170*, 56–63. https://doi.org/10.1016/j.biocon.2013.12.036

Fricke, R., Eschmeyer, W. N., & van der Laan, R. (2025). Eschmeyer's Catalog of Fishes: Genera, species, references. Eschmeyer's Catalog of Fishes. http://researcharchive.calacademy.org/research/ichthyology/catalog/fishcatmain.asp

Funk, W. C., McKay, J. K., Hohenlohe, P. A., & Allendorf, F. W. (2012). Harnessing genomics for delineating conservation units. *Trends in Ecology & Evolution*, *27*(9), 489–496. https://doi.org/10.1016/j.tree.2012.05.012

Garrett, G. P., Hubbs, C., & Edwards, R. J. (2002). Threatened fishes of the world: *Cyprinodon pecosensis* Echelle & Echelle, 1978 (Cyprinodontidae). *Environmental Biology of Fishes*, *65*(3), 366. https://doi.org/10.1023/A:1020521230163

Gilpin, M. E., & Soulé, M. E. (1986). Minimum viable populations: Processes of species extinction. In M. E. Soulé (Ed.), *Conservation biology: The science of scarcity and diversity* (pp. 19–34). Sinauer.

Girard, C. F. (1856). Researches upon the cyprinoid fishes inhabiting the fresh waters of the United States, west of the Mississippi Valley, from specimens in the museum of the Smithsonian Institution. *Proceedings of the Academy of Natural Sciences of Philadelphia*, *8*, 165–213.

Girard, C. F. (1859a). Ichthyological notices. *Proceedings of the Academy of Natural Sciences of Philadelphia*, *11*(60-77), 157–161.

Girard, C. F. (1859b). Ichthyological notices. *Proceedings of the Academy of Natural Sciences of Philadelphia*,*11*, 56–68.

Goudet, J., & Jombart, T. (2022). *Hierfstat: Estimation and Tests of Hierarchical F-Statistics* (Version Package Version 0.5-7). https://CRAN.R-project.org/package=hierfstat

Gruber, B., Unmack, P. J., Berry, O. F., & Georges, A. (2018). dartR: An R package to facilitate analysis of SNP data generated from reduced representation genome sequencing. *Molecular Ecology Resources*, *18*(3), 691–699. https://doi.org/10.1111/1755-0998.12745

Harley, G. L., & Maxwell, J. T. (2018). Current declines of Pecos River (New Mexico, USA) streamflow in a 700-year context. *The Holocene*, *28*(5), 767–777. https://doi.org/10.1177/0959683617744263

Hedrick, P. W., Adams, J. R., & Vucetich, J. A. (2011). Reevaluating and broadening the definition of genetic rescue. *Conservation Biology*, *25*, 1069–1070.

Hedrick, P. W., Robinson, J. A., Peterson, R. O., & Vucetich, J. A. (2019). Genetics and extinction and the example of isle Royale wolves. *Animal Conservation*, *22*(3), 302–309. https://doi.org/10.1111/acv.12479

Hendrickson, D. A., & Cohen, A. E. (2022). *Fishes of Texas Project Database (Version 3.0)* (US). https://doi.org/10.17603/C3WC70

Hoagstrom, C. W., & Brooks, J. E. (1999). *Distribution, status, and conservation of the Pecos pupfish, cyprinodon pecosensis*. Technical Report No. 2 (pp. 1–76). New Mexico Department of Game and Fish.

Hoagstrom, C. W., Davenport, S. R., & Osborne, M. J. (2025). Assembling the Pecos River fish fauna: Barrier displacement on the southern Great Plains, North America. *Biological Reviews*, *100*(4), 1534–1556. https://doi.org/10.1111/brv.70012

Hoagstrom, C. W., & Osborne, M. J. (2021). Biogeography of *Cyprinodon* across the Great Plains-Chihuahuan Desert region and adjacent areas. In *Proceedings of the desert fishes council special publication* (pp. 20–76). Desert Fishes Council.

Hoffmann, A. A., Sgrò, C. M., & Kristensen, T. N. (2017). Revisiting adaptive potential, population size, and conservation. *Trends in Ecology & Evolution*, *32*(7), 506–517. https://doi.org/10.1016/j.tree.2017.03.012

Hothorn, T., Bretz, F., & Westfall, P. (2008). Simultaneous inference in general parametric models. *Biometrical Journal*, *50*(3), 346–363. https://doi.org/10.1002/bimj.200810425

Jelks, H. L., Walsh, S. J., Burkhead, N. M., Contreras-Balderas, C., Díaz-Pardo, E., Hendrickson, D. A., Lyons, J., Mandrak, N. E., McCormick, F., Nelson, J. S., Platania, S. P., Porter, B. A., Renaud, C. B., Jacobo Schmitter-Soto, J., Taylor, E. B., & Warren, M. L., Jr. (2008). Conservation status of imperiled north American freshwater and diadromous fishes. *Fisheries*, *33*(8), 372–407.

Jensen, R., Hatler, W., Mecke, M., & Hart, C. (2006). The Influences of Human Activities on the Waters of the Pecos Basin of Texas: A Brief Overview. Texas Water Resources Institute. https://hdl.handle.net/1969.1/94964

Johnson, K. (1996). Gypsum karst in the United States. *International Journal of Speleology*, *25*(3/4), 183–193. https://doi.org/10.5038/1827-806X.25.3.13

Jombart, T. (2008). *Adegenet*: A R package for the multivariate analysis of genetic markers. *Bioinformatics*, *24*(11), 1403–1405. https://doi.org/10.1093/bioinformatics/btn129

Kalogianni, E., Koutsikos, N., Smeti, E., Kapakos, Y., & Vardakas, L. (2025). Evaluating the feasibility and effectiveness of removing the globally invasive eastern mosquitofish *Gambusia holbrooki* from lotic environments. *Environmental Management*, *75*(3), 477–489. https://doi.org/10.1007/s00267-024-02054-5

Kalra, A., Joshi, N., Baral, S., Nhuchhen Pradhan, S., Mambepa, M., Paudel, S., Xia, C., & Gupta, R. (2021). Coupled 1D and 2D HEC-RAS floodplain modeling of pecos river in New Mexico. In *World Environmental and Water Resources Congress 2021* (pp. 165–178). American Society of Civil Engineers. https://doi.org/10.1061/9780784483466.016

Kodric-Brown, A., & Mazzolini, P. (1992). The breeding system of pupfish, *Cyprinodon pecosensis*: Effects of density and interspecific interactions with the killifish, Fundulus zebrinus. *Environmental Biology of Fishes*, *35*(2), 169–176. https://doi.org/10.1007/bf00002191

Kodric-Brown, A., & Rosenfield, J. A. (2004). Populations of Pecos pupfish (*Cyprinodon pecosensis*) differ in their susceptibility to hybridization with sheepshead minnow (*C. Variegatus*). *Behavioral Ecology and Sociobiology*, *56*(2), 116–123. https://doi.org/10.1007/s00265-004-0761-1

Krueger, C. C., & May, B. (1991). Ecological and genetic effects of salmonid introductions in North America. *Canadian Journal of Fisheries and Aquatic Sciences*, *48*(S1), 66–77. https://doi.org/10.1139/f91-305

Lacepède, B. G. E. (1803). *Histoire naturelle des poissons* (Vol. 5, pp. 1–803). Paris: F. Dufart.

Lawson, D. J., van Dorp, L., & Falush, D. (2018). A tutorial on how not to over-interpret STRUCTURE and ADMIXTURE bar plots. *Nature Communications*, *9*(1), 3258. https://doi.org/10.1038/s41467-018-05257-7

Lee, D. S., Gilbert, C. R., Hocutt, C. H., Jenkins, R. E., McAllister, D. E., & Stauffer, J. R. (1980). *Atlas of north American freshwater fishes*. North Carolina State Museum of Natural History.

Maddison, W. P. (1997). Gene trees in species trees. *Systematic Biology*, *46*(3), 523–536.

Martin, C. H., Crawford, J. E., Turner, B. J., & Simons, L. H. (2016). Diabolical survival in Death Valley: Recent pupfish colonization, gene flow and genetic assimilation in the smallest species range on earth. *Proceedings of the Royal Society B: Biological Sciences*, *283*(1823), 20152334. https://doi.org/10.1098/rspb.2015.2334

McCracken, G. F., Parker, C. R., & Guffey, S. Z. (1993). Genetic differentiation and hybridization between stocked hatchery and native Brook trout in Great Smoky Mountains National Park. *Transactions of the American Fisheries Society*, *122*(4), 533–542. https://doi.org/10.1577/1548-8659(1993)122%253C0533:GDAHBS%253E2.3.CO;2

Miller, R. R. (1961). Man and the changing fish fauna of the American southwest. *Michigan Academy of Science, Arts, and Letters*, *46*, 365–404.

Minckley, W. L., Meffe, G. K., & Soltz, D. L. (1991). Conservation and Management of Short-lived Fishes: The cyprinodontoids. In W. L. Minckley & J. E. Deacon (Eds.), *Battle against extinction: Native fish management in the American west* (pp. 247–282). University of Arizona Press.

Moyle, P. B. (1976). Fish introductions in California: History and impact on native fishes. *Biological Conservation*, *9*(2), 101–118. https://doi.org/10.1016/0006-3207(76)90043-4

Moyle, P. B., & Leidy, R. A. (1992). Loss of biodiversity in aquatic ecosystems: Evidence from fish faunas. In *Conservation biology: The theory and practice of nature conservation preservation and management* (pp. 127–169). Springer. https://doi.org/10.1007/978-1-4684-6426-9_6

Nei, M., Maruyama, T., & Chakraborty, R. (1975). The bottleneck effect and genetic variability in populations. *Evolution*, *29*(1), 1–10.

Olden, J. D., & Poff, N. L. (2005). Long-term trends of native and non-native fish faunas in the American southwest. *Animal Biodiversity and Conservation*, *28*(1), 75–89. doi:10.32800/abc.2005.28.0075

O'Leary, S. J., Puritz, J. B., Willis, S. C., Hollenbeck, C. M., & Portnoy, D. S. (2018). These aren't the loci you're looking for: Principles of effective SNP filtering for molecular ecologists. *Molecular Ecology*, *27*(16), 3193–3206. https://doi.org/10.1111/mec.14792

Osborne, M. J., Portnoy, D. S., Fields, A. T., Bean, M. G., Hoagstrom, C. W., & Conway, K. W. (2021). Under the radar: Genetic assessment of Rio Grande shiner (*Notropis jemezanus*) and speckled chub (*Macrhybopsis aestivalis*), two Rio Grande basin endemic cyprinids that have experienced recent range contractions. *Conservation Genetics*, *22*(2), 187–204. https://doi.org/10.1007/s10592-020-01328-9

Page, L. M., & Burr, B. M. (2011). *Peterson Field Guide to Freshwater Fishes of North America North of Mexico, Second*. Houghton Mifflin Harcourt.

Pavlova, A., Beheregaray, L. B., Coleman, R., Gilligan, D., Harrisson, K. A., Ingram, B. A., … Sunnucks, P. (2017). Severe consequences of habitat fragmentation on genetic diversity of an endangered Australian freshwater fish: A call for assisted gene flow. *Evolutionary Applications*, *10*, 531–550.

Perkin, J., Troia, M., & Acre, M. (2021). Conservation status of native fishes in the Chihuahuan Desert region of the United States: A spatial perspective. In *Proceedings of the desert fishes council special publication* (pp. 77–103). Desert Fishes Council. doi:10.26153/tsw/21665

Perry, W. L., Lodge, D. M., & Feder, J. L. (2002). Importance of hybridization between indigenous and nonindigenous freshwater species: An overlooked threat to north American biodiversity. *Systematic Biology*, *51*(2), 255–275. https://doi.org/10.1080/10635150252899761

Peterson, B. K., Weber, J. N., Kay, E. H., Fisher, H. S., & Hoekstra, H. E. (2012). Double digest RADseq: An inexpensive method for De novo SNP discovery and genotyping in model and non-model species. *PLoS One*, *7*(5), 11.

Philipp, D. P., Childers, W. F., & Whitt, G. S. (1983). A biochemical genetic evaluation of the northern and Florida subspecies of largemouth bass. *Transactions of the American Fisheries Society*, *112*(1), 1–20. https://doi.org/10.1577/1548-8659(1983)112%253C1:ABGEOT%253E2.0.CO;2

Portnoy, D. S., Bretzing-Tungate, R. J., Fields, A. T., Bean, M. G., Smith, R. K., Dolan, E. P., Blanchard, R., & Conway, K. W. (2025). A total evidence approach justifies taxonomic splitting of the endangered Pecos gambusia into three species. *Royal Society Open Science*, *12*(11), 251025. https://doi.org/10.1098/rsos.251025

Puritz, J. B., Hollenbeck, C. M., & Gold, J. R. (2014). *dDocent*: A RADseq, variant-calling pipeline designed for population genomics of non-model organisms. *PeerJ*, *2*, e431. https://doi.org/10.7717/peerj.431

Pyke, G. H. (2008). Plague minnow or mosquito fish? A review of the biology and impacts of introduced gambusia species. *Annual Review of Ecology, Evolution, and Systematics*, *39*, 171–191. https://doi.org/10.1146/annurev.ecolsys.39.110707.173451

R Core Team. (2019). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing.

Roberts, J. H., Angermeier, P. L., & Hallerman, E. M. (2013). Distance, dams and drift: What structures populations of an endangered, benthic

stream fish? *Freshwater Biology*, *58*(10), 2050–2064. https://doi.org/10.1111/fwb.12190
Rosenfield, J. A., & Kodric-Brown, A. (2003). Sexual selection promotes hybridization between Pecos pupfish, *Cyprinodon pecosensis* and sheepshead minnow, *C. Variegatus*. *Journal of Evolutionary Biology*, *16*(4), 595–606. https://doi.org/10.1046/j.1420-9101.2003.00557.x
Rosenfield, J. A., Nolasco, S., Lindauer, S., Sandoval, C., & Kodric-Brown, A. (2004). The role of hybrid vigor in the replacement of Pecos pupfish by its hybrids with sheepshead minnow. *Conservation Biology*, *18*(6), 1589–1598. https://doi.org/10.1111/j.1523-1739.2004.00356.x
Schönhuth, S., Hillis, D. M., Neely, D. A., Lozano-Vilano, L., Perdices, A., & Mayden, R. L. (2012). Phylogeny, diversity, and species delimitation of the north American round-nosed minnows (Teleostei: Dionda), as inferred from mitochondrial and nuclear DNA sequences. *Molecular Phylogenetics and Evolution*, *62*(1), 427–446.
Serve, N. (2013). *Cyprinodon pecosensis*. The IUCN red list of threatened species 2013: E.T6163A15361876. https://doi.org/10.2305/IUCN.UK.2013-1.RLTS.T6163A15361876.en
Smith, G. R., Badgley, C., Eiting, T. P., & Larson, P. S. (2010). Species diversity gradients in relation to geological history in north American freshwater fishes. *Evolutionary Ecology Research*, *12*(6), 693–726.
Smith, R. J., Kulp, M. A., & Fitzpatrick, B. M. (2025). Barriers Broken: Genetic swamping in restored Brook Trout populations. *PLOS One*, *20*(12), e0327926. https://doi.org/10.1371/journal.pone.0327926
Twardek, W. M., Taylor, J. J., Rytwinski, T., Aitken, S. N., MacDonald, A. L., Van Bogaert, R., & Cooke, S. J. (2023). The application of assisted migration as a climate change adaptation tactic: An evidence map and synthesis. *Biological Conservation*, *280*, 109932. https://doi.org/10.1016/j.biocon.2023.109932
U.S. Fish and Wildlife Service. (2024). *Endangered and Threatened Wildlife and Plants; Threatened Species Status With Section 4(d) Rule for Pecos Pupfish and Designation of Critical Habitat* (89 Fed. Reg. 92744–92785) [Proposed rule]. https://www.federalregister.gov/documents/2024/11/22/2024-27127/endangered-and-threatened-wildlife-and-plants-threatened-species-status-with-section-4d-rule-for
Vähä, J.-P., & Primmer, C. R. (2006). Efficiency of model-based Bayesian methods for detecting hybrid individuals under different hybridization scenarios and with different numbers of loci. *Molecular Ecology*, *15*(1), 63–72. https://doi.org/10.1111/j.1365-294X.2005.02773.x
Waples, R. K., Larson, W. A., & Waples, R. S. (2016). Estimating contemporary effective population size in non-model species using linkage disequilibrium across thousands of loci. *Heredity*, *117*(4), 233–240.
Warren, M. L., Jr., & Burr, B. M. (2015). *Freshwater fishes of North America: Petromyzontidae to Catostomidae* (Vol. 1). John Hopkins University Press. https://doi.org/10.1007/s10641-014-0372-2
Weir, B. S., & Cockerham, C. C. (1984). Estimating F-statistics for the analysis of population structure. *Evolution*, *38*, 1358–1370. https://doi.org/10.1111/j.1558-5646.1984.tb05657.x
Whitlock, M. C., & Lotterhos, K. E. (2015). Reliable detection of loci responsible for local adaptation: Inference of a null model through trimming the distribution of F(ST). *The American Naturalist*, *186*(Suppl 1), S24–S36. https://doi.org/10.1086/682949
Whitmore, D. H. (1983). Introgressive hybridization of smallmouth bass (*Micropterus dolomieui*) and Guadalupe bass (*M. treculi*). *Copeia*, *1983*(3), 672. https://doi.org/10.2307/1444331
Wilcox, J. L., & Martin, A. P. (2006). The devil's in the details: Genetic and phenotypic divergence between artificial and native populations of the endangered pupfish (*Cyprinodon diabolis*). *Animal Conservation*, *9*(3), 316–321.
Wilde, G. R., & Echelle, A. A. (1992). Genetic status of Pecos pupfish populations after establishment of a hybrid swarm involving an introduced congener. *Transactions of the American Fisheries Society*, *121*(3), 277–286. https://doi.org/10.1577/1548-8659(1992)121%253C0277:gsoppp%253E2.3.co;2
Willi, Y., Buskirk, J. V., & Hoffmann, A. A. (2006). Limits to the adaptive potential of small populations. *Annual Review of Ecology, Evolution, and Systematics*, *37*, 433–458. https://doi.org/10.1146/annurev.ecolsys.37.091305.110145
Willis, S. C., Hollenbeck, C. M., Puritz, J. B., Gold, J. R., & Portnoy, D. S. (2017). Haplotyping RAD loci: An efficient method to filter paralogs and account for physical linkage. *Molecular Ecology Resources*, *17*(5), 955–965. https://doi.org/10.1111/1755-0998.12647
Wright, S. (1931). Evolution in mendelian populations. *Genetics*, *16*(2), 1–97.
Xu, Q. (2017). *Body shape diversification of Pecos pupfish (Cyprinodon Pecosensis) on varying habitats as evaluated by geometric morphometrics [masters]*. Western Kentucky University.
Yuan, F., Miyamoto, S., & Anand, S. (2007). Changes in major element hydrochemistry of the Pecos River in the American southwest since 1935. *Applied Geochemistry*, *22*(8), 1798–1813. https://doi.org/10.1016/j.apgeochem.2007.03.036

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Dolan, E. P., Fields, A. T., Bean, M. G., Doege, R. R., Hatt, J. L., Conway, K. W., & Portnoy, D. S. (2026). Conservation genomics of Pecos pupfish (*Cyprinodon pecosensis*). *Journal of Fish Biology*, 1–15. https://doi.org/10.1111/jfb.70486